Cloud Resource Usage Analysis for Cost-Aware Application Deployment

Student Research Proposal • Abidur Rahman Niloy • Cloud Computing • Data Analysis • Software Infrastructure

Proposed by	Abidur Rahman Niloy
Project type	Student-led research / prototype study
Estimated duration	8–10 weeks

Project Overview

I want to explore how basic usage data can be analyzed to understand cloud-resource consumption and identify simple opportunities for cost-aware deployment. The project will use simulated or permitted usage records rather than connecting to a real billing account, making it suitable for a student research environment.

Student perspective
I want to keep the first version realistic for a student team: a focused research question, a reproducible experiment, and a prototype that can be discussed with a researcher and improved through collaboration.

Research Problem

Cloud platforms make it easy to scale applications, but students and small teams may not understand how resource choices affect usage and cost. A research prototype can examine resource-utilization patterns and create simple recommendations without requiring access to private billing information.

Research Objectives

- Create a small, realistic dataset of application resource usage.
- Analyze CPU, memory, storage, and request-related patterns.
- Develop simple rules or statistical indicators for inefficient resource allocation.
- Build a dashboard that explains usage patterns in an understandable way.
- Evaluate whether the recommendations remain useful under different simulated workloads.

Research Questions

RQ1. Which resource metrics are most useful for identifying over-provisioning?

RQ2. Can simple usage thresholds provide useful cost-awareness signals?

RQ3. How do workload patterns affect resource recommendations?

RQ4. How can a dashboard explain infrastructure trade-offs to beginners?

Proposed Methodology

Step 1. Design a simulated or permission-based cloud workload dataset.

Step 2. Clean and aggregate resource measurements over time.

Step 3. Explore relationships between workload intensity and resource utilization.

Step 4. Create baseline threshold-based recommendations.

Step 5. Test the recommendations under several synthetic workload scenarios.

Step 6. Build visual summaries and document limitations of simulated cost estimates.

Data / Materials / Experimental Resources

Synthetic workload logs, optional cloud free-tier resources, local database, dashboard interface, and documented assumptions about pricing.

Evaluation Plan

Compare recommendation consistency across workload scenarios and measure whether simulated over-provisioning is reduced without causing unacceptable resource shortages.

Expected Challenges

- Cloud pricing differs by provider, region, and service.
- Simulated workloads cannot represent every real deployment.
- Recommendations may become outdated as usage patterns change.
- Resource savings should not be considered separately from reliability.

Reproducibility and Documentation

I will keep the main assumptions, dataset versions, preprocessing decisions, experiment settings, and evaluation results documented. The aim is that another student can understand how the result was produced and repeat the core experiment without relying on undocumented manual steps.

Planned Mitigation

I will begin with a small baseline and a feasibility check. If the original scope becomes too large for the available time or resources, I will narrow the research question rather than adding unnecessary complexity. Limitations and failed experiments will be documented as part of the research process.

Expected Outcomes

- A resource-usage analysis dataset and pipeline.
- A simple cost-awareness dashboard.
- Comparison of recommendation rules across workloads.
- A discussion of the trade-off between efficiency, cost, and reliability.

Potential Research Collaboration

I am looking for someone interested in cloud systems, data analysis, or infrastructure who can help design realistic workloads, interpret resource metrics, or improve the dashboard.

Technical Stack

Python, SQL, JavaScript, Node.js, optional AWS services, Git.

Scope, Ethics, and Responsible Research

No private cloud billing or personal account information is required. If a cloud provider is used, only authorized resources and free-tier or explicitly permitted usage will be included in the experiment.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, feasibility check
2. Setup	Data/system preparation and baseline implementation
3. Experiments	Development, comparison, tuning, and documentation
4. Evaluation	Metrics, error analysis, limitations, reproducibility
5. Reporting	Results, discussion, documentation, and presentation

What I Am Looking for in a Collaborator

I am looking for someone who is genuinely interested in the research question and willing to contribute to discussion, experiment design, implementation, analysis, or writing. I would like to divide responsibilities according to each person's strengths and keep the work transparent and documented.

Possible Next Step

If a researcher is interested, I would first like to discuss feasibility, refine the research question, and agree on a realistic division of work. The proposal can then be updated into a more specific study plan with milestones.

Note: This is a proposed student project. It does not claim completed research or guaranteed results, and the scope can be refined through collaboration.